

Charles Oliphant

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EDUCATION

Brigham Young University

April 2027

B.S. in Computer Science, Machine Learning Emphasis

Provo, UT

Relevant Coursework: CS 473 Advanced Machine Learning, CS 470 Introduction to Artificial Intelligence, CS 270 Introduction to Machine Learning, MATH 380 Mathematics of Statistics, MATH 213 Linear Algebra

EXPERIENCE

Concepts NREC

April 2022 - Present

Engineering Intern

- Develop and maintain Python data-processing and visualization software used weekly to analyze hundreds of pump tests, compare results, validate performance, detect faults, and generate reports and Excel workbooks.
- Built a pump design prototype using Python, NumPy, and SciPy that generates initial pump geometry for early-stage design; scheduled for internal deployment and commercial release in 2026.
- Developed a PyTorch feed-forward neural network trained on processed CFD data to predict compressor exit-flow parameters, achieving an R-squared of approximately 0.90 on test data; planned for integration into design tools.
- Provisioned a two-node Kubernetes cluster with Ansible and two NVIDIA RTX A5000 GPUs to serve Qwen 27B using vLLM for an internal AI platform.
- Built Neo4j GraphRAG and SQL-backed RAG systems over approximately 1,800 HTML documentation files, enabling documentation-grounded answers to software-use questions.

Brigham Young University

Fall 2023 - Winter 2026

Teaching Assistant, CS 111 Introduction to Computer Science

- Led Python programming labs for 10-30 students, reinforcing foundational programming and problem-solving concepts.
- Provided one-on-one help in a support lab, debugging code and explaining programming assignments.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

Machine Learning and Data: PyTorch, scikit-learn, pandas, NumPy, SciPy, Jupyter, RAG, Neo4j

Infrastructure: Kubernetes, Ansible, vLLM, Git, Linux

Engineering: CFD, sensor-data processing, data visualization, Excel

LEADERSHIP AND SERVICE

The Church of Jesus Christ of Latter-day Saints

December 2019 - November 2021

Volunteer Missionary and District Leader

Philippines and Newport Beach, CA

- Led and trained a district of 6-8 missionaries, tracking goals, teaching progress, and individual development.

Eagle Scout

2018